

Comparative study of capillary zone electrophoresis and micellar electrokinetic capillary chromatography for the separation of naphthalenedisulfonate isomers

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We investigated the separation of a test mixture of nine substituted and unsubstituted naphthalenedisulfonate isomers by capillary electrophoresis with a UV diode array detector. In particular, we focused on how the composition of the running buffer affected the separation selectivity. When capillary zone electrophoresis was carried out, the best results were obtained when organic solvents such as ethanol or propan-2-ol were added. Eight peaks were baseline separated but in no case were all the unsubstituted isomers separated. Therefore, capillary electrophoretic separation of the compounds was examined in the presence of micellar agents, such as sodium dodecyl sulfate, polyethylene glycol dodecyl ether (Brij 35) and cetyltrimethylammonium bromide. All the substituted isomers and two of the unsubstituted isomers were well resolved within 20 min by micellar electrokinetic capillary chromatography when Brij 35 was used as micellar agent. Separations were reproducible, in terms of peak area and migration time, under these conditions.

Introduction

Naphthalenesulfonate isomers are used in many industrial processes, particularly in the production of plasticizers for concrete, tanning agents and dyes. These substances are surface-active and can be modified by microorganisms.^{1,2} Little is known about their possible effects on the environment or about their toxicity and biodegradation.³

Measurements in industrial sewage and several European river waters have demonstrated that these compounds occur in the aquatic environment and cannot be totally removed in mechanical–biological waste water treatment plants. They remain in the water owing to their high polarity.^{4–6} Some of them have even been detected in drinking waters,^{6,7} which means that they are potentially hazardous for surface and groundwater and are of concern with regard to drinking water quality.

Many analytical methods have been proposed for separating and determining naphthalenesulfonate isomers. Gas chromatography (GC) has been used, although it requires a tedious derivatization step and several problems occur in the derivatization of isomers which also contain amino and hydroxy groups.^{7,8} High performance liquid chromatography (HPLC)^{7,9,10} and ion pair chromatography (IPC)^{11–13} are certainly the most commonly used techniques. Nowadays, capillary electrophoresis (CE) is a rapidly growing, powerful technique that is preferred in many applications to conventional chromatographic techniques because it is efficient, flexible, cheap to operate and has short analysis times.

Although CE seems a promising technique for the determination of naphthalenesulfonate isomers, few reports have been published so far. It has been used to separate 21 naphthalenemono- and -disulfonate isomers in two runs and some factors affecting resolution have been studied.¹⁴ More recently, a comparison between capillary zone electrophoresis (CZE) and micellar electrokinetic capillary chromatography (MEKC) showed that MEKC gave better selectivity in the separation of 11 naphthalenemonosulfonates.¹⁵ Because of their polarity and the considerable variety of isomers, naphthalenesulfonates are difficult to determine and several workers have attempted to

separate test mixtures by adding additives such as surfactants¹⁶ or cyclodextrins.¹⁷ In spite of this, the separation was still incomplete in some cases. Some reports indicate that HPLC, IPC and CE are suitable and complementary techniques for the determination of naphthalenesulfonate isomers.^{17–19}

In this paper, we describe and compare several CE separation strategies for the resolution of an isomeric naphthalenedisulfonate (NDS) test mixture. On the basis of previous results, we studied the use of different types of additives such as organic modifiers and surfactants to improve the separation further. We also discuss the influence of the separation conditions.

Experimental

Instrumentation

All separations were performed with a Hewlett-Packard (Palo Alto, CA, USA) ³DCE system equipped with a UV diode array detector. Uncoated fused silica capillaries (Supelco, Bellefonte, PA, USA), and 64.5 cm × 75 µm id, were used throughout. A window was created 8.5 cm from the end of the capillary and 230 nm was selected as the absorbing wavelength. The capillary was thermostatted using a Peltier system at the temperatures described in the text. Hydrodynamic injections were made at 50 mbar for 10 s, except for the cetyltrimethylammonium bromide (CTAB) micellar system, where electrokinetic injection was made by applying –4 kV for 10 s. The system was used with either positive or reversed polarity (CTAB micellar system). Electropherograms were analysed using an HP Chemstation chromatographic data system (version A:04:02).

Reagents and chemicals

The NDS isomers used in this study (see Table 1) were purchased from Fluka (Buchs, Switzerland), except 2,7-NDS and 7-NH₂-1,3-NDS, which were obtained from Acros Organics (Geel, Belgium) and Aldrich (Beerse, Belgium), re-

spectively. A stock standard solution of 1000 mg l⁻¹ of each compound was prepared in water (several drops of NaOH were usually added to enhance solubility) and stored in a dark-glass flask at 4 °C. Working standard solutions containing 5 mg l⁻¹ of each compound were prepared daily by diluting the stock standard solutions with purified water.

Sodium tetraborate and polyethylene glycol dodecyl ether (Brij 35) were obtained from Fluka. Sodium dodecyl sulfate (SDS) and CTAB were supplied by Sigma (St. Louis, MO, USA). Methanol, ethanol, propan-2-ol and acetonitrile were of HPLC grade from Merck (Darmstadt, Germany). Background electrolytes (BGEs) were prepared daily, de-gassed by ultrasonication and filtered through 0.45 µm nylon syringe filters (Tracer, Teknokroma, Barcelona, Spain) prior to use. All standards and BGEs were prepared using Milli-Q water (Millipore, Bedford, MA, USA).

For the SDS micellar system, the capillary had to be conditioned between runs by flushing 0.1 M NaOH for 1 min, followed by water for 2 min and running buffer for 3 min. Using this method, it was found that the run-to-run reproducibility was maintained. For the other systems, the capillary was conditioned between runs by flushing it only with the BGE for 3 min. Each surfactant was studied using a new capillary. Capillaries were stored in water when not in use.

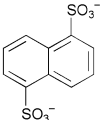
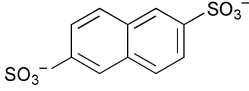
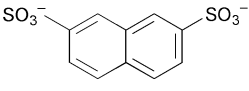
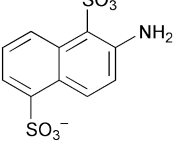
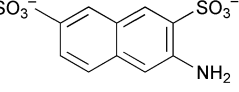
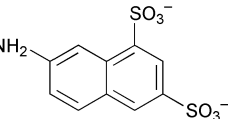
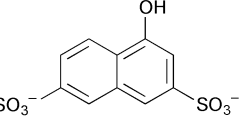
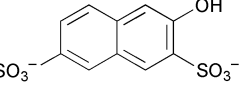
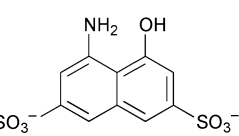
Results and discussion

CZE method development

Although naphthalenesulfonates are strong acids, completely dissociated in aqueous or aqueous–organic solutions over a broad pH range, aminonaphthalenesulfonates are not and their acid–base properties are similar to those of carboxylic acids. Therefore, compounds carrying an amino group are expected to be strongly influenced by pH. The running buffers were 20 mM borate buffers at different pH values. After several experiments, pH 9.1 was selected as the optimum pH value for the separation of the NDS isomers in a relatively short time. At pH > 9.5 the resolution improved slightly but it also had a negative effect on migration time and peak shapes.

The influence of the buffer concentration on the migration time and resolution of nine NDS compounds was investigated from 10 to 30 mM sodium borate buffer at pH 9.1. At 30 mM borate buffer, all substituted NDS isomers were resolved but the migration time also increased undesirably owing to the increase in ionic strength. Therefore, 20 mM [Fig. 1(a)] was chosen as the concentration which struck the best compromise between resolution and migration time. The three isomeric unsubstituted NDS were not resolved at any borate concentration.

Table 1 Naphthalenedisulfonate isomers studied with their abbreviations and peak assignments

Compound	Abbreviation	Structures	Peak assignment
Naphthalene-1,5-disulfonate	1,5-NDS		4
Naphthalene-2,6-disulfonate	2,6-NDS		5
Naphthalene-2,7-disulfonate	2,7-NDS		6
2-Aminonaphthalene-1,5-disulfonate	2-NH ₂ -1,5-NDS		3
3-Aminonaphthalene-2,7-disulfonate	3-NH ₂ -2,7-NDS		2
7-Aminonaphthalene-1,3-disulfonate	7-NH ₂ -1,3-NDS		1
1-Hydroxynaphthalene-3,6-disulfonate	1-OH-3,6-NDS		9
2-Hydroxynaphthalene-3,6-disulfonate	2-OH-3,6-NDS		7
8-Amino-1-hydroxynaphthalene-3,6-disulfonate	8-NH ₂ -1-OH-3,6-NDS		8

In principle, the ideal separation is obtained by applying the highest voltage possible because the plate number in CE is proportional to the voltage.²⁰ However, the maximum voltage is limited by the Joule heat generated during the separation process, which broadens the peaks and decreases the separation efficiency. At 30 kV, the analysis time was shorter and the currents were not excessive, so this voltage was selected as the optimum run voltage.

Organic modifiers can also be added to tune an electrophoretic separation. They can influence both the electroosmotic (EOF) flow and the electrophoretic mobilities. The decrease in EOF flow is mainly due to an increase in viscosity. The electrophoretic mobilities of the analytes are affected by the changes in both the charge and the hydrodynamic volume, and also by the type and percentage of organic modifier.²¹

We studied how adding various organic solvents (methanol, ethanol, propan-2-ol and acetonitrile) at various concentrations (up to 20% v/v) to the running buffer affected the electrophoretic behaviour of a test mixture of NDS. The migration time of the NDS increased as the content of organic solvents in the buffer was increased.

Since protic solvents (ethanol, propanol, *etc.*) are more efficient than aprotic solvents (acetonitrile) at increasing the viscosity and reducing the dielectric constant of the electrolyte, the former had a greater effect on migration times than the latter. The increased electrophoretic mobilities towards the injection end led to longer separation times and larger differences in mobilities and therefore to better peak resolution. However, the temperature had to be raised to 45 °C to shorten the analysis time when ethanol and propan-2-ol were used.

The electropherogram in Fig. 1(a) shows that most of the NDS studied could not be resolved by the buffer alone. No significant improvements were observed when methanol (re-

sults not shown) was added. However, using ethanol (12% v/v) or propan-2-ol (15% v/v), baseline separation was almost complete: only two isomeric unsubstituted NDS (peaks 5 and 6) co-eluted [Fig. 2(b) and (c)]. When propan-2-ol was used, the peak resolution was better but the analysis time increased significantly [Fig. 2(c)]. When 15% v/v acetonitrile (results not shown) was added, the results for the first group of eluted compounds were similar to those obtained with ethanol. The analysis time was shorter but peak 8 moved directly under peak 9. Selectivity was also affected by the nature of the organic solvent, and the migration order of some compounds was reversed.

Although CZE gave almost complete separation when organic modifiers (ethanol, propan-2-ol) were added to the borate buffer, this approach cannot be recommended because the separation conditions (30 kV and 45 °C) required to shorten the analysis time worsened the repeatability between runs because the Joule heat was greater.

MEKC method development

Sometimes the selectivity of a CE separation can also be improved by adding micellar agents, *i.e.*, by performing MEKC. Since the selectivity is influenced considerably by the type and concentration of the micellar agents, several types of surfactants were investigated.²² Adding organic modifiers to such systems was also evaluated.

SDS is an anionic surfactant which has a relatively strong hydrogen-bond-donating character; this means that it tends to incorporate analytes which accept hydrogen bonds, so the selectivity for such compounds can be increased. Varying the concentration of the surfactant between 25 and 100 mM

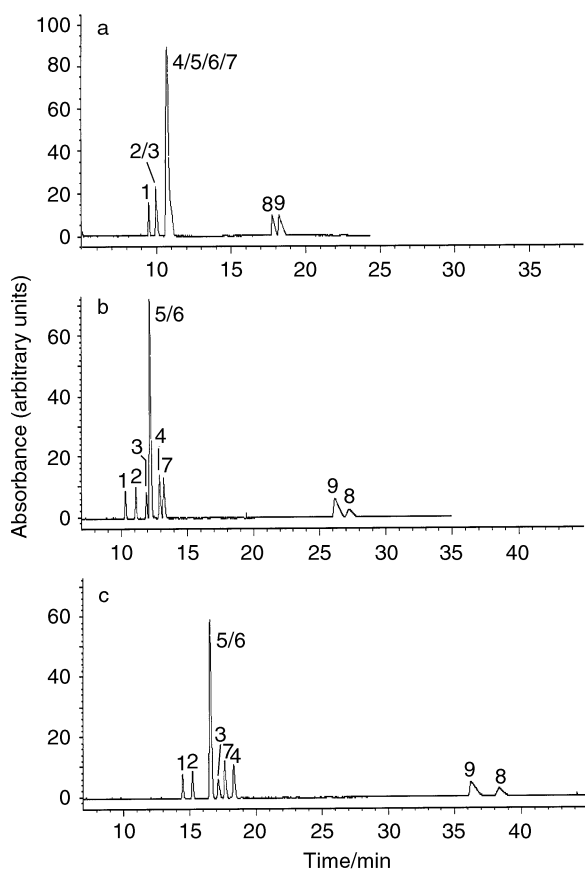


Fig. 1 Separation of an NDS test mixture by CZE in a pure aqueous (a) 20 mM borate buffer (pH 9.1) at 30 kV and 25 °C and in a mixed organic-aqueous 20 mM borate buffer at 30 kV and 45 °C with (b) 12% v/v ethanol and (c) 15% v/v propan-2-ol. Peak identification as in Table 1.

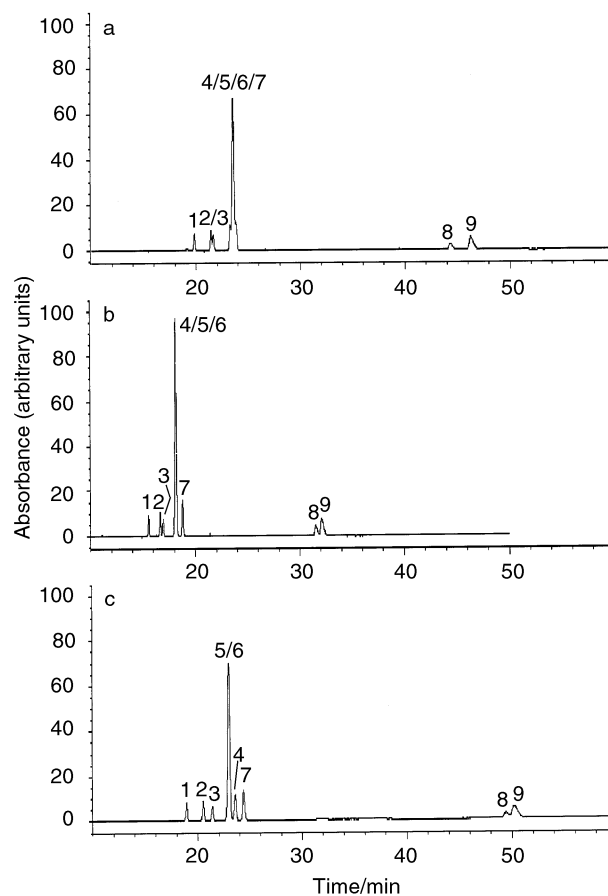


Fig. 2 Separation of an NDS test mixture by MEKC using 20 mM borate buffer and 75 mM SDS at 19 kV and (a) 25 °C, (b) 40 °C and (c) with 5% v/v acetonitrile at 40 °C. Peak identification as in Table 1.

primarily causes a considerable increase in the migration time. This is due to the fact that anionic micelles are electrostatically attracted towards the anode but, because of the prevalent velocity of the EOF, they slowly migrate towards the cathode, *i.e.*, in the direction of the detector. Consequently, SDS micelles selectively decrease the migration time of the analytes with which they interact by partitioning mechanisms between the aqueous phase and the micellar phase.

As shown in Fig. 2(a), addition of SDS to the 20 mM borate buffer at 25 °C did not improve the resolution of NDS because of the strong interactions between the analytes and the SDS micelles. The thermal effects on the separation efficiency in MEKC are well-known. An increase in temperature will decrease the concentration of the free micelles because the critical micellar concentration (c.m.c.) increases. This affects the partition coefficient and can either improve or impair MEKC separations.²³ The influence of temperature is illustrated in Fig. 2. Increasing the temperature from 25 °C [Fig. 2(a)] to 40 °C [Fig. 2(b)] resulted in different electrophoretic mobilities of the analytes, which significantly improved the resolution. It also decreased the total analysis time from 48 to 33 min. Unfortunately, the three isomeric unsubstituted NDS could not be resolved.

Finally, we studied the possibility of further improving the separation by simultaneously adding SDS and organic modifiers (ethanol, propan-2-ol and acetonitrile) to the running buffer. Of the organic modifiers studied, only acetonitrile enhanced the resolution of the test mixture. As shown in Fig. 2(c), adding 5% v/v acetonitrile to the micellar buffer made it possible to separate 1,5-NDS (peak 4), but the separation of 2,6- and 2,7-NDS (peaks 5 and 6) was still not satisfactory. The migration times of the compounds that eluted last were also unacceptably long, even when 30 kV was applied at 40 °C across the capillary during the analysis.

Brij 35 is a non-ionic surfactant which is a strong hydrogen-bond base and a weak hydrogen-bond acid, and it has a great capacity for attraction interactions between ion pairs. On the basis of previous results which showed the usefulness of adding Brij 35 to separate naphthalenemonosulfonates,¹⁵ we used a similar buffer for the present mixture of NDS. The influence of the Brij 35 concentration on the electrophoretic behaviour of the selected NDS was investigated between 10 and 60 mM. The results indicate that eight NDS isomers can be well resolved within 22 min when a 20 mM borate buffer containing 25 mM Brij 35 is used (Fig. 3). Brij 35 is also more attractive than SDS because separation can be optimum at ambient temperature

(25 °C), thus preventing Joule heat effects, and because the capillary does not need to be flushed with NaOH between runs for separations to be reproducible. Despite these good results, the separation of 2,6- and 2,7-NDS (peaks 5 and 6) was still not satisfactory. When Brij 35 was used as the surfactant, then, the addition of organic modifiers only led to increased migration times.

Taking into account that cationic surfactants such as CTAB are potentially very useful because electrostatic interactions between anionic analytes and cationic micelles lead to high separation selectivities, the separation of the three unsubstituted isomers was finally evaluated using CTAB as micellar agent. When a BGE containing CTAB at high enough concentrations is used, the charge of the capillary wall is reversed, which in turn reverses the EOF.²⁴ When this happens, the EOF migrates towards the anodic end of the capillary and the polarity of the power supply must be reversed to let the anionic compounds reach the detection window.

The concentration of CTAB in the BGE was increased from 2 to 10 mM, but when no organic modifiers were added to the BGE, only closely eluting, broad peaks were obtained. As can be seen from Fig. 4, the three unsubstituted isomers (peaks 4, 5 and 6), which could not be resolved in either free solution or in SDS- or Brij 35 MEKC systems, were successfully resolved by adding 7 mM CTAB and 34% v/v of propan-2-ol to the borate buffer. Although CTAB is used above its c.m.c., it does not form micelles, because a high content of propan-2-ol was added to the buffer. Because of this, resolution was improved by additional ion-pair interactions.

In conclusion, MEKC can be considered to be a promising technique for an essentially complete separation of even unsubstituted NDS. Except for the inability to resolve two of the three unsubstituted NDS included in this study, Brij 35 MEKC is an efficient, fast and easy to handle method for separating complex mixtures of NDS isomers.

Validation of the method

To test the performance of the Brij 35 MEKC method, we studied the linearity and precision with standard solution mixtures under the optimum conditions described above. Calibration graphs (five points) were constructed for standard solutions between 0.5 and 25 mg l⁻¹; the linearity was good ($r^2 > 0.997$) for all the compounds. Limits of quantification (LOQ)

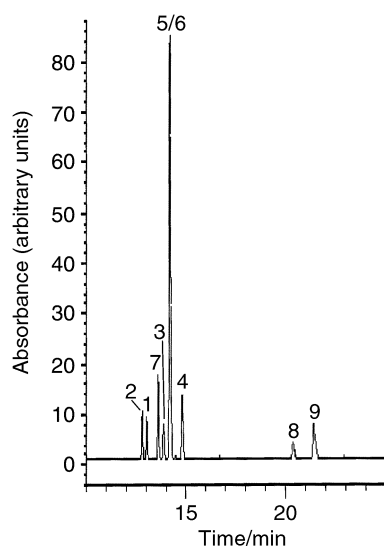


Fig. 3 Separation of an NDS test mixture by MEKC using 20 mM borate buffer and 25 mM Brij 35 at 20 kV and 25 °C. Peak identification as in Table 1.

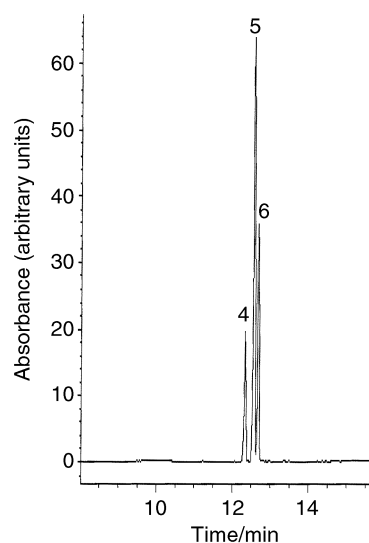


Fig. 4 Separation of unsubstituted NDS using 20 mM borate, 7 mM CTAB and 34% v/v propan-2-ol at 30 kV (negative power supply) and 25 °C. Peak identification as in Table 1.

Table 2 LOQ, LOD and repeatability data (RSD) for NDS isomers separated by Brij 35 MEKC

Compound	LOD/ mg l ⁻¹	LOQ/ mg l ⁻¹	RSD (%) ^a	
			Migration times	Normalized peak areas
7-NH ₂ -1,3-NDS	0.2	0.5	0.2	1.1
3-NH ₂ -2,7-NDS	0.2	0.5	0.2	2.2
2-NH ₂ -1,5-NDS	0.2	0.5	0.2	1.1
1,5-NDS	0.1	0.5	0.5	1.3
2,6-NDS	0.1	0.5	0.2	2.7
2-OH-3,6-NDS	0.15	0.5	0.3	2.3
8-NH ₂ -1-OH- 3,6-NDS	0.15	0.5	0.3	3.1
1-OH-3,6-NDS	0.15	0.5	0.3	4.3

^a RSD was determined from five replicates.

and limits of detection (LOD) based on a signal-to-noise ratio of 10 and 3, respectively were determined from five repetitions for standard solutions (Table 2). The precision of the method was evaluated by determining its repeatability and reproducibility. The repeatability values obtained for five successive injections of standard solution containing 0.2 mg l⁻¹ of each NDS varied between 0.18 and 0.51% for migration times and 1.06 and 4.28 for normalized peak areas (Table 2). Reproducibility values were determined in the same way as repeatability data but on different days. It was observed that the day-to-day precision values were similar to the within-day precision values.

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